

Supplementary Information

Baseline-source sensitivity and robustness of supplied bias-corrected CMIP6 daily precipitation-extreme projections for Uttaradit Province, Thailand

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Scope and evidence boundary

This Supplementary Information reports only tables and diagnostics derived from the previously audited outputs used in the main manuscript. No new climate-model simulation, bias-correction run, projection window, bootstrap analysis, model weighting, or spatial-independence analysis was added during revision.

Table S1 provides all 66 scenario–period–index combinations; Tables S2–S3 expand the historical performance and residual-bias summaries; Supplementary Note S1 records why archived trend p-values were excluded. Missing bias-correction provenance, source-grid metadata, model genealogy, and author-owned declarations are not inferred.

Complete baseline-source sensitivity

Table S1a. Complete baseline-source sensitivity results for SSP2-4.5 (33 combinations).

Scenario-period	Index	Model-consistent signed native-unit change, median [Q1, Q3]	Model-consistent change, median [Q1, Q3] (%)	Model agreement; class	Observed-reference signed native-unit change, median [Q1, Q3]	Observed-reference change, median [Q1, Q3] (%)	Observed agreement; class	Baseline effect (obs – model)
SSP2-4.5 2021–2040	PRCPTOT	+6.5 [-60.5, +92.4]	+0.6 [-5.3, +9.6]%	4/7; ambiguous	-47.1 [-65.2, -34.2]	-4.2 [-5.8, -3.1]%	6/7; robust decrease	-4.8 pp; sign reversal: yes; class changed: yes
SSP2-4.5 2021–2040	SDII	-0.0 [-0.2, +0.1]	-0.3 [-1.3, +1.1]%	4/7; ambiguous	+1.8 [+1.7, +2.1]	+17.8 [+16.7, +20.1]%	7/7; robust increase	+18.1 pp; sign reversal: yes; class changed: yes
SSP2-4.5 2021–2040	Rx1day	+7.7 [-1.6, +11.6]	+11.4 [-2.2, +16.0]%	4/7; ambiguous	-2.1 [-5.1, +4.4]	-2.7 [-6.6, +5.7]%	4/7; ambiguous	-14.1 pp; sign reversal: yes; class changed: no
SSP2-4.5 2021–2040	Rx5day	+4.9 [-5.6, +19.9]	+2.8 [-3.5, +12.3]%	4/7; ambiguous	+25.7 [+19.0, +43.2]	+17.8 [+13.2, +30.0]%	7/7; robust increase	+15.1 pp; sign reversal: no; class changed: yes
SSP2-4.5 2021–2040	CDD	-0.6 [-1.2, +0.2]	-0.9 [-1.9, +0.4]%	5/7; ambiguous	+13.4 [+12.0, +15.5]	+28.1 [+25.2, +32.5]%	7/7; robust increase	+29.0 pp; sign reversal: yes; class changed: yes
SSP2-4.5 2021–2040	CWD	+0.3 [-0.2, +0.6]	+2.9 [-2.0, +5.1]%	5/7; ambiguous	-20.8 [-21.4, -20.2]	-64.7 [-66.5, -62.9]%	7/7; robust decrease	-67.6 pp; sign reversal: yes; class changed: yes
SSP2-4.5 2021–2040	R10mm	-1.2 [-2.9, +1.5]	-3.6 [-7.9, +4.8]%	4/7; ambiguous	+2.6 [+2.5, +3.3]	+8.6 [+8.2, +10.7]%	7/7; robust increase	+12.2 pp; sign reversal: yes; class changed: yes
SSP2-4.5 2021–2040	R20mm	-1.1 [-1.4, +1.0]	-6.8 [-8.9, +7.5]%	4/7; ambiguous	+2.7 [+2.4, +3.3]	+22.8 [+20.2, +27.3]%	7/7; robust increase	+29.5 pp; sign reversal: yes; class changed: yes
SSP2-4.5 2021–2040	R50mm†	+0.1 [+0.0, +0.4]	+7.0 [+0.3, +21.6]%	5/7; ambiguous	+0.1 [+0.1, +0.3]	+6.1 [+2.5, +11.9]%	6/7; robust increase	-0.9 pp; sign reversal: no; class changed: yes
SSP2-4.5 2021–2040	R95p	+19.0 [-16.8, +35.3]	+5.2 [-4.4, +11.3]%	4/7; ambiguous	+69.2 [+63.0, +86.3]	+23.3 [+21.2, +29.0]%	7/7; robust increase	+18.0 pp; sign reversal: no; class changed: yes
SSP2-4.5 2021–2040	R99p	+11.4 [+1.6, +26.8]	+8.4 [+1.1, +21.0]%	5/7; ambiguous	+37.0 [+32.6, +48.8]	+31.5 [+27.8, +41.6]%	7/7; robust increase	+23.1 pp; sign reversal: no; class changed: yes
SSP2-4.5 2041–2060	PRCPTOT	+39.7 [-18.0, +65.6]	+3.5 [-1.8, +6.0]%	4/7; ambiguous	+12.1 [-108.8, +67.8]	+1.1 [-9.7, +6.1]%	4/7; ambiguous	-2.4 pp; sign reversal: no; class changed: no
SSP2-4.5 2041–2060	SDII	+0.2 [-0.1, +1.1]	+1.3 [-0.7, +9.1]%	5/7; ambiguous	+2.3 [+1.5, +3.3]	+22.1 [+14.1, +32.0]%	6/7; robust increase	+20.8 pp; sign reversal: no; class changed: yes
SSP2-4.5 2041–2060	Rx1day	+11.0 [+0.8, +15.1]	+17.1 [+1.2, +20.1]%	5/7; ambiguous	-0.8 [-4.5, +12.9]	-1.1 [-5.9, +16.7]%	4/7; ambiguous	-18.1 pp; sign reversal: yes; class changed: no
SSP2-4.5 2041–2060	Rx5day	+21.0 [+12.7, +46.4]	+12.5 [+7.5, +27.4]%	6/7; robust increase	+53.0 [+28.2, +73.4]	+36.8 [+19.6, +51.0]%	6/7; robust increase	+24.3 pp; sign reversal: no; class changed: no
SSP2-4.5 2041–2060	CDD	-1.1 [-3.3, +1.4]	-1.9 [-5.2, +2.2]%	4/7; ambiguous	+14.0 [+10.7, +15.0]	+29.5 [+22.6, +31.4]%	7/7; robust increase	+31.3 pp; sign reversal: yes; class changed: yes
SSP2-4.5 2041–2060	CWD	-0.2 [-0.3, +0.9]	-2.0 [-2.9, +8.9]%	4/7; ambiguous	-20.7 [-21.3, -20.4]	-64.4 [-66.1, -63.5]%	7/7; robust decrease	-62.4 pp; sign reversal: no; class changed: yes
SSP2-4.5 2041–2060	R10mm	-1.3 [-2.0, +0.3]	-4.4 [-5.3, +0.7]%	4/7; ambiguous	+3.9 [+0.7, +4.5]	+12.7 [+2.3, +14.7]%	5/7; ambiguous	+17.2 pp; sign reversal: yes; class changed: no
SSP2-4.5 2041–2060	R20mm	-0.3 [-0.5, +0.9]	-2.2 [-3.0, +5.8]%	4/7; ambiguous	+4.2 [+1.8, +4.5]	+35.0 [+15.0, +37.7]%	6/7; robust increase	+37.2 pp; sign reversal: yes; class changed: yes
SSP2-4.5 2041–2060	R50mm†	+0.6 [+0.2, +1.1]	+30.8 [+8.4, +45.3]%	6/7; robust increase	+0.6 [+0.0, +1.2]	+27.7 [+2.2, +57.5]%	5/7; ambiguous	-3.0 pp; sign reversal: no; class changed: yes
SSP2-4.5 2041–2060	R95p	+36.1 [+5.7, +106.9]	+9.9 [+1.5, +27.6]%	6/7; robust increase	+110.7 [+52.1, +197.1]	+37.2 [+17.5, +66.3]%	6/7; robust increase	+27.3 pp; sign reversal: no; class changed: no
SSP2-4.5 2041–2060	R99p	+37.6 [+11.6, +87.4]	+26.5 [+8.4, +55.4]%	6/7; robust increase	+62.0 [+32.9, +127.7]	+52.8 [+28.0, +108.7]%	6/7; robust increase	+26.2 pp; sign reversal: no; class changed: no
SSP2-4.5 2081–2100	PRCPTOT	-38.7 [-132.2, +14.3]	-3.4 [-12.7, +1.7]%	5/7; ambiguous	-68.2 [-209.5, -19.1]	-6.1 [-18.7, -1.7]%	6/7; robust decrease	-2.7 pp; sign reversal: no; class changed: yes

Scenario–period	Index	Model-consistent signed native-unit change, median [Q1, Q3]	Model-consistent change, median [Q1, Q3] (%)	Model agreement; class	Observed-reference signed native-unit change, median [Q1, Q3]	Observed-reference change, median [Q1, Q3] (%)	Observed agreement; class	Baseline effect (obs – model)
SSP2-4.5 2081–2100	SDII	+0.1 [-1.0, +0.2]	+0.8 [-8.4, +1.9]%	4/7; ambiguous	+1.7 [+0.7, +2.3]	+16.4 [+6.6, +22.3]%	6/7; robust increase	+15.6 pp; sign reversal: no; class changed: yes
SSP2-4.5 2081–2100	Rx1day	+4.9 [-3.6, +11.5]	+6.2 [-5.7, +16.6]%	5/7; ambiguous	+1.3 [-11.1, +7.6]	+1.6 [-14.4, +9.9]%	4/7; ambiguous	-4.5 pp; sign reversal: no; class changed: no
SSP2-4.5 2081–2100	Rx5day	+9.0 [-8.7, +31.0]	+6.0 [-5.6, +18.4]%	5/7; ambiguous	+31.3 [+5.1, +58.0]	+21.7 [+3.5, +40.3]%	5/7; ambiguous	+15.7 pp; sign reversal: no; class changed: no
SSP2-4.5 2081–2100	CDD	-2.0 [-2.3, +0.3]	-3.2 [-3.7, +0.5]%	5/7; ambiguous	+12.9 [+11.7, +14.6]	+27.0 [+24.6, +30.7]%	7/7; robust increase	+30.2 pp; sign reversal: yes; class changed: yes
SSP2-4.5 2081–2100	CWD	-0.1 [-0.7, +0.6]	-1.1 [-5.8, +5.0]%	4/7; ambiguous	-20.7 [-21.8, -19.9]	-64.5 [-67.7, -61.9]%	7/7; robust decrease	-63.4 pp; sign reversal: no; class changed: yes
SSP2-4.5 2081–2100	R10mm	-3.8 [-5.1, -0.9]	-10.3 [-14.8, -2.2]%	5/7; ambiguous	+1.5 [-1.9, +2.6]	+4.8 [-6.3, +8.5]%	4/7; ambiguous	+15.1 pp; sign reversal: yes; class changed: no
SSP2-4.5 2081–2100	R20mm	-0.7 [-2.9, +0.2]	-4.3 [-19.1, +1.4]%	5/7; ambiguous	+2.4 [-0.0, +3.8]	+20.1 [-0.4, +31.9]%	5/7; ambiguous	+24.4 pp; sign reversal: yes; class changed: no
SSP2-4.5 2081–2100	R50mm†	+0.3 [-0.3, +0.6]	+18.1 [-14.0, +27.8]%	5/7; ambiguous	+0.4 [-0.5, +0.8]	+18.4 [-24.4, +35.6]%	4/7; ambiguous	+0.3 pp; sign reversal: no; class changed: no
SSP2-4.5 2081–2100	R95p	+24.5 [-48.3, +51.8]	+8.2 [-14.5, +13.4]%	4/7; ambiguous	+98.2 [-17.2, +141.9]	+33.0 [-5.8, +47.7]%	5/7; ambiguous	+24.8 pp; sign reversal: no; class changed: no
SSP2-4.5 2081–2100	R99p	+28.7 [-10.8, +47.8]	+24.7 [-9.6, +30.2]%	5/7; ambiguous	+72.0 [-2.1, +88.0]	+61.3 [-1.8, +74.9]%	5/7; ambiguous	+36.6 pp; sign reversal: no; class changed: no

Note: Model-consistent = future supplied bias-corrected minus supplied bias-corrected historical 1995–2014; observed-reference = future supplied bias-corrected minus observed 1995–2014. Values are medians and Q1–Q3 across seven model-specific regional contrasts after equal weighting of 13 stations. “Signed native-unit change” is not an absolute magnitude. Agreement is the number of models sharing the ensemble direction; $\geq 6/7$ is a descriptive robustness screen, not a significance test. Baseline effect is the signed observed-reference median minus the model-consistent median in percentage points. †R50mm percentages are denominator-sensitive; signed days yr⁻¹ should be prioritized.

Table S1b. Complete baseline-source sensitivity results for SSP5-8.5 (33 combinations).

Scenario–period	Index	Model-consistent signed native-unit change, median [Q1, Q3]	Model-consistent change, median [Q1, Q3] (%)	Model agreement; class	Observed-reference signed native-unit change, median [Q1, Q3]	Observed-reference change, median [Q1, Q3] (%)	Observed agreement; class	Baseline effect (obs – model)
SSP5-8.5 2021–2040	PRCPTOT	+32.4 [-10.9, +58.1]	+3.1 [-1.3, +5.5]%	5/7; ambiguous	-27.4 [-131.5, +11.9]	-2.4 [-11.7, +1.1]%	5/7; ambiguous	-5.5 pp; sign reversal: yes; class changed: no
SSP5-8.5 2021–2040	SDII	-0.1 [-0.4, +0.4]	-0.8 [-2.9, +3.4]%	4/7; ambiguous	+1.6 [+1.3, +2.5]	+15.9 [+12.8, +24.1]%	7/7; robust increase	+16.7 pp; sign reversal: yes; class changed: yes
SSP5-8.5 2021–2040	Rx1day	+8.8 [-0.1, +12.0]	+12.5 [-0.0, +15.8]%	5/7; ambiguous	+2.5 [-6.0, +8.2]	+3.3 [-7.8, +10.7]%	4/7; ambiguous	-9.2 pp; sign reversal: no; class changed: no
SSP5-8.5 2021–2040	Rx5day	+9.6 [-0.6, +42.2]	+6.4 [-0.5, +25.5]%	5/7; ambiguous	+38.8 [+17.6, +63.8]	+26.9 [+12.2, +44.3]%	7/7; robust increase	+20.5 pp; sign reversal: no; class changed: yes
SSP5-8.5 2021–2040	CDD	-0.2 [-2.3, +1.5]	-0.3 [-3.7, +2.3]%	4/7; ambiguous	+13.9 [+11.1, +14.1]	+29.1 [+23.3, +29.6]%	7/7; robust increase	+29.4 pp; sign reversal: yes; class changed: yes
SSP5-8.5 2021–2040	CWD	+0.3 [+0.1, +0.5]	+2.2 [+0.8, +4.8]%	6/7; robust increase	-20.2 [-21.1, -19.9]	-62.8 [-65.7, -61.9]%	7/7; robust decrease	-65.0 pp; sign reversal: yes; class changed: yes
SSP5-8.5 2021–2040	R10mm	-0.1 [-1.1, +0.3]	-0.2 [-3.4, +1.0]%	4/7; ambiguous	+2.1 [+0.7, +4.8]	+6.8 [+2.4, +15.7]%	5/7; ambiguous	+7.1 pp; sign reversal: yes; class changed: no
SSP5-8.5 2021–2040	R20mm	+0.1 [-1.3, +0.6]	+0.4 [-8.4, +4.5]%	4/7; ambiguous	+3.2 [+1.3, +3.9]	+26.6 [+11.1, +32.4]%	6/7; robust increase	+26.2 pp; sign reversal: no; class changed: yes
SSP5-8.5 2021–2040	R50mm†	+0.3 [-0.0, +0.7]	+16.5 [-1.5, +29.7]%	5/7; ambiguous	+0.2 [-0.2, +0.7]	+7.7 [-8.5, +34.2]%	4/7; ambiguous	-8.8 pp; sign reversal: no; class changed: no
SSP5-8.5 2021–2040	R95p	+23.4 [-22.8, +70.7]	+7.8 [-6.9, +17.9]%	5/7; ambiguous	+67.9 [+26.0, +141.5]	+22.8 [+8.7, +47.6]%	6/7; robust increase	+15.0 pp; sign reversal: no; class changed: yes

Scenario–period	Index	Model-consistent signed native-unit change, median [Q1, Q3]	Model-consistent change, median [Q1, Q3] (%)	Model agreement; class	Observed-reference signed native-unit change, median [Q1, Q3]	Observed-reference change, median [Q1, Q3] (%)	Observed agreement; class	Baseline effect (obs – model)
SSP5-8.5 2021–2040	R99p	+28.0 [-6.5, +58.5]	+24.1 [-4.9, +36.9]%	5/7; ambiguous	+31.0 [+21.0, +91.3]	+26.4 [+17.9, +77.7]%	6/7; robust increase	+2.3 pp; sign reversal: no; class changed: yes
SSP5-8.5 2041–2060	PRCPTOT	-17.2 [-127.6, -8.6]	-1.9 [-11.6, -0.8]%	6/7; robust decrease	-116.5 [-195.2, +6.2]	-10.4 [-17.4, +0.6]%	5/7; ambiguous	-8.6 pp; sign reversal: no; class changed: yes
SSP5-8.5 2041–2060	SDII	-0.7 [-1.5, -0.5]	-5.5 [-12.6, -3.7]%	6/7; robust decrease	+1.4 [+0.0, +1.8]	+13.8 [+0.1, +16.9]%	5/7; ambiguous	+19.3 pp; sign reversal: yes; class changed: yes
SSP5-8.5 2041–2060	Rx1day	+1.5 [-0.2, +6.6]	+2.0 [-0.4, +8.9]%	5/7; ambiguous	+1.1 [-11.4, +4.5]	+1.5 [-14.8, +5.8]%	4/7; ambiguous	-0.5 pp; sign reversal: no; class changed: no
SSP5-8.5 2041–2060	Rx5day	+7.6 [-11.5, +12.5]	+4.2 [-7.4, +7.6]%	4/7; ambiguous	+33.3 [+1.2, +38.5]	+23.2 [+0.9, +26.8]%	5/7; ambiguous	+18.9 pp; sign reversal: no; class changed: no
SSP5-8.5 2041–2060	CDD	-2.6 [-3.5, -0.5]	-4.2 [-5.6, -0.8]%	5/7; ambiguous	+10.5 [+10.3, +13.6]	+22.0 [+21.7, +28.7]%	7/7; robust increase	+26.1 pp; sign reversal: yes; class changed: yes
SSP5-8.5 2041–2060	CWD	+0.6 [+0.0, +0.7]	+4.8 [+0.5, +7.1]%	5/7; ambiguous	-20.2 [-21.0, -19.9]	-63.0 [-65.2, -61.9]%	7/7; robust decrease	-67.8 pp; sign reversal: yes; class changed: yes
SSP5-8.5 2041–2060	R10mm	-1.7 [-5.0, -0.9]	-4.5 [-15.3, -2.7]%	6/7; robust decrease	-0.2 [-1.2, +4.6]	-0.7 [-4.0, +14.8]%	4/7; ambiguous	+3.8 pp; sign reversal: no; class changed: yes
SSP5-8.5 2041–2060	R20mm	-1.2 [-3.3, -0.8]	-9.4 [-21.7, -5.1]%	6/7; robust decrease	+1.3 [-0.5, +3.7]	+11.1 [-4.2, +30.7]%	4/7; ambiguous	+20.5 pp; sign reversal: yes; class changed: yes
SSP5-8.5 2041–2060	R50mm†	-0.1 [-0.4, +0.2]	-6.4 [-22.7, +9.7]%	4/7; ambiguous	+0.0 [-0.7, +0.4]	+1.8 [-34.7, +19.5]%	4/7; ambiguous	+8.2 pp; sign reversal: yes; class changed: no
SSP5-8.5 2041–2060	R95p	-28.2 [-68.0, +11.4]	-7.3 [-21.1, +2.9]%	4/7; ambiguous	+60.3 [-43.1, +107.8]	+20.3 [-14.5, +36.2]%	4/7; ambiguous	+27.6 pp; sign reversal: yes; class changed: no
SSP5-8.5 2041–2060	R99p	+2.9 [-26.7, +19.6]	+1.8 [-20.1, +12.5]%	4/7; ambiguous	+46.4 [-17.0, +60.2]	+39.5 [-14.5, +51.2]%	4/7; ambiguous	+37.7 pp; sign reversal: no; class changed: no
SSP5-8.5 2081–2100	PRCPTOT	-492.5 [-524.9, -189.2]	-46.7 [-52.8, -17.3]%	7/7; robust decrease	-513.0 [-636.6, -220.9]	-45.8 [-56.9, -19.7]%	7/7; robust decrease	+0.9 pp; sign reversal: no; class changed: no
SSP5-8.5 2081–2100	SDII	-3.4 [-4.4, -2.0]	-27.7 [-37.3, -15.9]%	7/7; robust decrease	-1.4 [-3.0, +0.1]	-13.1 [-28.8, +0.7]%	5/7; ambiguous	+14.6 pp; sign reversal: no; class changed: yes
SSP5-8.5 2081–2100	Rx1day	-20.2 [-21.4, -6.5]	-25.6 [-31.7, -8.5]%	6/7; robust decrease	-18.4 [-31.1, -7.8]	-23.9 [-40.4, -10.1]%	7/7; robust decrease	+1.7 pp; sign reversal: no; class changed: no
SSP5-8.5 2081–2100	Rx5day	-45.0 [-59.8, -13.9]	-25.2 [-39.0, -8.3]%	6/7; robust decrease	-10.3 [-50.2, +8.7]	-7.1 [-34.9, +6.0]%	4/7; ambiguous	+18.0 pp; sign reversal: no; class changed: yes
SSP5-8.5 2081–2100	CDD	+0.1 [-2.4, +3.5]	+0.2 [-3.8, +5.6]%	4/7; ambiguous	+13.3 [+11.4, +18.9]	+27.9 [+24.0, +39.7]%	7/7; robust increase	+27.7 pp; sign reversal: no; class changed: yes
SSP5-8.5 2081–2100	CWD	-1.7 [-2.5, -0.6]	-14.0 [-21.8, -6.2]%	6/7; robust decrease	-22.7 [-23.4, -21.0]	-70.8 [-72.9, -65.2]%	7/7; robust decrease	-56.8 pp; sign reversal: no; class changed: no
SSP5-8.5 2081–2100	R10mm	-18.1 [-18.5, -7.9]	-51.1 [-59.0, -22.3]%	7/7; robust decrease	-12.6 [-17.6, -3.4]	-40.9 [-57.5, -11.0]%	7/7; robust decrease	+10.2 pp; sign reversal: no; class changed: no
SSP5-8.5 2081–2100	R20mm	-8.8 [-9.9, -4.6]	-62.2 [-69.1, -29.7]%	7/7; robust decrease	-5.9 [-7.8, -1.1]	-49.3 [-65.3, -8.9]%	5/7; ambiguous	+12.8 pp; sign reversal: no; class changed: yes
SSP5-8.5 2081–2100	R50mm†	-1.2 [-1.5, -0.6]	-68.6 [-72.3, -25.9]%	6/7; robust decrease	-1.4 [-1.6, -0.5]	-65.8 [-76.6, -25.5]%	6/7; robust decrease	+2.8 pp; sign reversal: no; class changed: no
SSP5-8.5 2081–2100	R95p	-205.9 [-241.4, -101.8]	-64.5 [-69.8, -27.7]%	7/7; robust decrease	-160.5 [-202.4, -30.4]	-54.0 [-68.0, -10.2]%	6/7; robust decrease	+10.6 pp; sign reversal: no; class changed: no
SSP5-8.5 2081–2100	R99p	-76.3 [-99.3, -33.8]	-62.6 [-69.0, -22.8]%	6/7; robust decrease	-57.2 [-78.7, -4.1]	-48.7 [-67.0, -3.5]%	5/7; ambiguous	+13.9 pp; sign reversal: no; class changed: yes

Note: Model-consistent = future supplied bias-corrected minus supplied bias-corrected historical 1995–2014; observed-reference = future supplied bias-corrected minus observed 1995–2014. Values are medians and Q1–Q3 across seven model-specific regional contrasts after equal weighting of 13 stations. “Signed native-unit change” is not an absolute magnitude. Agreement is the number of models sharing the ensemble direction; $\geq 6/7$ is a descriptive

robustness screen, not a significance test. Baseline effect is the signed observed-reference median minus the model-consistent median in percentage points. †R50mm percentages are denominator-sensitive; signed days yr⁻¹ should be prioritized.

Extended historical evaluation

Table S2a. Historical performance across 91 model–station pairs, reported as median [Q1, Q3].

Scale	Series	N	r	RMSE	MAE	PBIAS (%)	NSE	KGE
Daily	Raw	91	0.037 [0.017, 0.063]	10.18 [7.94, 10.87]	4.02 [3.53, 4.67]	-21.9 [-59.0, -0.1]	-0.353 [-0.525, -0.190]	-0.120 [-0.261, -0.041]
Daily	Supplied bias-corrected	91	0.057 [0.042, 0.074]	11.84 [9.39, 13.53]	4.87 [4.17, 5.05]	-9.0 [-16.6, 7.8]	-0.968 [-1.339, -0.822]	0.012 [-0.061, 0.045]
Monthly	Raw	91	0.272 [0.154, 0.374]	103.98 [93.46, 112.59]	73.51 [67.07, 80.09]	-21.9 [-59.0, -0.1]	-0.308 [-0.556, -0.126]	0.162 [-0.139, 0.277]
Monthly	Supplied bias-corrected	91	0.441 [0.340, 0.492]	108.73 [86.33, 123.07]	69.59 [57.83, 79.24]	-9.0 [-16.6, 7.8]	-0.344 [-0.589, -0.161]	0.385 [0.267, 0.446]

Note: Daily RMSE and MAE are in mm d⁻¹; monthly RMSE and MAE are in mm month⁻¹. Daily records contain 12,410 or 12,418 model-specific days; monthly records contain 408 months. Daily and monthly PBIAS derive from the same accumulated rainfall and are not independent evidence.

Table S2b. Paired model–station changes after correction relative to raw series.

Scale	Lower RMSE	Lower PBIAS	Higher r	Higher KGE
Daily	8/91 (8.8%)	61/91 (67.0%)	72/91 (79.1%)	78/91 (85.7%)
Monthly	40/91 (44.0%)	61/91 (67.0%)	74/91 (81.3%)	78/91 (85.7%)

Note: These fractions are paired correction-versus-raw comparisons. Improvement in one metric does not imply overall improvement; RMSE increased for most daily pairs, and median NSE remained negative.

Table S3. Residual historical bias of supplied corrected series relative to observations, 1995–2014.

Index	Unit	Median signed regional difference	Median regional bias (%)	Q1–Q3 regional bias (%)	Maximum absolute regional bias (%)
PRCPTOT	mm yr ⁻¹	-59.84	-5.3	-10.1 to +0.8	17.3
SDII	mm wet-day ⁻¹	+1.87	+18.0	+16.0 to +20.6	22.0
Rx1day	mm	-6.32	-8.2	-12.6 to -1.7	16.3
Rx5day	mm	+21.18	+14.7	+11.6 to +20.1	24.1
CDD	days	+14.12	+29.6	+27.0 to +31.1	35.0
CWD	days	-20.60	-64.1	-67.1 to -62.9	70.2
R10mm	days yr ⁻¹	+3.72	+12.1	+5.5 to +19.2	21.4
R20mm	days yr ⁻¹	+3.58	+29.8	+18.8 to +35.0	39.6
R50mm	days yr ⁻¹	-0.05	-2.3	-8.6 to +8.4	21.8
R95p	mm yr ⁻¹	+65.68	+22.1	+12.7 to +30.3	33.9
R99p	mm yr ⁻¹	+24.34	+20.7	+14.3 to +34.3	37.5

Note: For each model, 1995–2014 annual index means were calculated by station, the 13 stations were equally averaged to a regional value, and corrected-minus-observed bias was then calculated. The table summarizes these seven model-level regional biases. The signed regional difference is expressed in the index's native unit; it is not an absolute magnitude. The maximum absolute percentage is across seven regional model values, not across stations.

Supplementary Note S1. Trend-test null diagnostic

The supplied trend module included a diagnostic self-test under AR(1) dependence. Its TFPW-MK false-positive fraction was 0.383 at nominal $\alpha = 0.05$. An independent audit of the previously supplied implementation used 2,000 AR(1) null series of length 20 with lag-1 coefficient 0.6 and seed 20260829; all three empirical false-positive fractions remained well above 0.05 (Table S4). These results falsify confirmatory type-I error control for the archived trend p-values, which were therefore excluded from the manuscript.

Table S4. Empirical false-positive fractions from supplied and independent null diagnostics.

Audit source	Test	n	AR(1) ϕ	Replicates	Seed	Nominal α	False-positive fraction
Supplied self-test	TFPW-MK	not recorded	0.6	not recorded	not recorded	0.05	0.3830
Independent audit	MK	20	0.6	2,000	20260829	0.05	0.2245
Independent audit	Hamed-Rao	20	0.6	2,000	20260829	0.05	0.2270
Independent audit	TFPW-MK	20	0.6	2,000	20260829	0.05	0.2665

Note: The archived JSON records n, ϕ , replicates, seed, α , and empirical fractions but not the innovation distribution, initialization/burn-in, sidedness, or all implementation details. This is therefore a diagnostic falsification/quality-assurance exercise rather than a fully reproducible Monte Carlo study.